BIT 324 Mobile Communication Network Design

Lesson One: Introduction Mobile Communication Network Design

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Lecture Overview

- **Computer Network**
- **Mobility of Bits & Bytes**
- **Wireless Computer Network**
- **Wireless-The Beginning**
- **Evolution of Wireless Networks**
- **Evolution of Wireless Data**
- **Evolution of Wireless LAN**
- **Evolution of Wireless PAN**
- **Mobile Computing**
- **Mobile Computing Functions**
- **Logical Functions of Mobile Computing**

Computer Network:

- A computer network is collection of several homogeneous/heterogeneous system, commonly used for resource sharing.
- Computer networks can be classified in many ways:
- Area wise:
  - LAN
  - MAN
  - WAN
- As per medium used:
  - Wired Computer Network
  - Wireless Computer Network

Wireless Computer Network:

A **Wireless Computer Network** is a type of network where computers and devices communicate with each other **without using physical cables**. Instead, they use **radio waves, infrared signals, or satellite signals** to transmit data.

A wireless computer network connects devices (computers, phones, printers, routers) using wireless communication technology such as **Wi-Fi or Bluetooth**.

Examples of Wireless Networks

WLAN (Wireless Local Area Network) – e.g., Wi-Fi in homes, schools, offices

WPAN (Wireless Personal Area Network) – e.g., Bluetooth

WMAN (Wireless Metropolitan Area Network) – e.g., city-wide WiMAX

WWAN (Wireless Wide Area Network) – e.g., mobile cellular networks (4G, 5G)

Components of a Wireless Network

- Wireless Router / Access Point
- Wireless Network Interface Card (NIC)
- Antennas
- End devices (laptops, smartphones, tablets)

Advantages of Wireless Network:

- **Flexible**: Radio waves can penetrate the obstacles. Sender and receiver can be placed any where.
- **Mobility**: Data can be access from any location.
- **Robustness** : Can survive in disaster (Earthquake, military operations)
- **Scalable**: Can be configured in variety of topologies.
- **Easy Installation**
- **Less Cost**
- **Usage of ISM band**: ISM (Industrial, Scientific and Medical) band (2.40GHz to 2.484 GHz, 5.725 GHz to 5.850 GHz) is available for use by anyone.
- **No Planning**: Only Wireless Ad hoc not required any planning.

Disadvantages of Wireless Network:

- **Quality of Service:**

✓Lower Bandwidth

✓Lower Data Transmission Rate

✓High Error Rates

✓Interference

✓Higher Delay

Restrictions: License-free frequency bands are not same worldwide.

- **Safety and Security**: Interference from other devices (e.g. Hospital. Eavesdropping is possible).

MOBILITY OF BITS AND BYTES

- In the last two centuries, mobility has been redefined.
- Both physical and virtual objects are now mobile.
  - Mobility of physical objects relate to movement of matters,
  - whereas movements of virtual objects relate to movements of bits and bytes.

MOBILITY OF BITS AND BYTES

- In the real world, objects move physically from one place to another.
- In the virtual world, nothing physically moves.
- The computer simply changes **data values (bits and bytes)** in memory.
- These changes update the object's position on the screen.
- Virtual movement = changing digital data.
- Physical movement = moving real objects**.**

MOBILITY OF BITS AND BYTES Cont.

- On October 4, 1957 the USSR (Union of Soviet Socialist Republic now mainly Russia) launched the Sputnik.
  - It was the first artificial earth satellite launched from Baikonur cosmodrome in Kazakhstan.
- In response to this, the US formed the Advanced Research Projects Agency (ARPA) within the Department of Defense (DoD).

MOBILITY OF BITS AND BYTES Cont.

- This laid the foundation of packet switched data networks.
- The important ones are TCP /IP and X.25.
  - TCP/I*P* was driven by education and defense in the USA
  - whereas X.25 was driven by European telecommunication industry and Governments.

WIRELESS-THE BEGINNING

- In 1947 researchers in AT&T Bell Labs conceived the idea of cellular phones.
- They realized that by using small service areas or cells they can reuse the frequency.
- This in turn can enhance the traffic capacity of mobile phones.

WIRELESS-THE BEGINNING Cont

- AT&T requested the Federal Communication Commission (FCC) to allocate a large number of radio-spectrum frequencies so that widespread mobile telephone service would become feasible.

Evolution of Wireless Networks

- The first wireless network was commissioned in Germany in 1958.
  - It was called A-Netz and used analog technology at 160 MHz.
  - Only outgoing calls were possible in this network.
  - That is to say that connection set-up was possible from the mobile station only.

Evolution of Wireless Networks Cont.

- This system evolved into B-Netz operating at the same 160 MHz.
  - It was possible to receive an incoming call from a fixed telephone network, provided that location of the mobile station was known.
- A-Netz was wireless but not a cellular network.
  - Therefore, these systems (A-Netz and B-Netz) did not have any function, which permitted handover or change of base station.

Evolution of Wireless Networks Cont.

- In 1968, in USA, the FCC reconsidered its position on Cellular network concept.
  - FCC agreed to allocate a larger frequency band for more number of mobile phones provided the technology to build a better mobile service be demonstrated.
  - AT&T and Bell Labs proposed a cellular system to the FCC with many small, low-powered, broadcast towers, each covering a hexagonal 'cell' of a few kilometers in radius.

Evolution of Wireless Networks Cont.

- Collectively these cells could cover a very large area.
- Each tower would use only a few of the total frequencies allocated to the system.
- As the phones traveled across the area, calls would be passed from tower to tower.

Evolution of Wireless Networks Cont.

- In April 1973, Martin Cooper of Motorola invented the first mobile phone handset and made the first call from a portable phone to Joel Engel.
- By 1977, AT&T and Bell Labs constructed a prototype of a public cellular network.
- In 1978, public trials of the cellular telephony system started in Chicago with over 2000 trial customers.

Evolution of Wireless Networks Cont.

- In 1982, FCC finally authorized commercial cellular service for the USA.
- A year later in 1983, the first American commercial analog cellular service AMPS (Advanced Mobile Phone Service) was made commercially available in Chicago.
  - This was the first cellular mobile network in the world

Evolution of Wireless Data

- Like the computers, the evolution of wireless technology has also been defined in generations.
- The first generation (1G) of wireless technology uses the analog technology.
  - It uses FDMA (Frequency Division Multiple Access) technology for modulation; for example, AMPS (Advanced Mobile Phone Service) in US.
  - The second generation or 2G technology uses digitized technology. It uses a combination of TDMA (Time Division Multiple Access) and FDMA technologies.
  - An example is GSM.

Evolution of Wireless Data Cont.

- In 2G technology, voice is digitized over a circuit.
- In 1G and 2G networks, data is transacted over circuits.
- This technology is called Circuit Switched Data or CSD in short.
- Using modems, a data connection is established between the device and the network.

Evolution of Wireless Data Cont.

- The next phase in the evolution is 2.5G.
  - In 2.5G technology, voice is digitized over a circuit.
  - However, data in 2.5G is packetized.
  - 2.5G uses the same encoding techniques as 2G does.
  - GPRS networks is an example of 2.5G.

Evolution of Wireless Data Cont.

- The Third Generation or 3G wireless technology makes a quantum leap from a technology point of view.
  - 3G uses Spread Spectrum techniques for media access and encoding.
  - In 3G networks, both data and voice use packets.
  - UMTS and CDMA2000 are examples of 3G networks.

Evolution of Wireless Data Cont.

- With the success of wireless telephony and messaging services like paging, wireless communication is beginning to be applied to the realm of personal and business computing in the domain of local area networks.
- Wireless LAN s are being deployed in homes, campuses, and commercial establishments.
- The domain of wireless data networks today comprises of Wireless PAN (Bluetooth, Infrared), Wireless LAN (IEEE 802.11 family) and Wireless WAN (Wide Area Networks), (GSM, GPRS, 3G)

Evolution of Wireless Data Cont.

Evolution of 4G (Fourth Generation Wireless Technology)

Developed to improve speed and performance beyond 3G networks

Officially launched around 2009–2010

Based on IP (Internet Protocol) technology for faster data transmission

Introduced LTE (Long Term Evolution) as the main 4G standard

Enabled high-speed internet, HD video streaming, online gaming, and video calls

Improved bandwidth, lower latency, and better network capacity

Laid the foundation for the development of 5G technology

In summary:4G evolved to provide faster, more reliable mobile internet and support modern digital services.

Evolution of Wireless Data Cont.

Evolution of 5G (Fifth Generation Wireless Technology)

Developed to overcome limitations of 4G networks

Commercial rollout began around 2019

Designed for ultra-high speed, low latency, and massive connectivity

Uses advanced technologies like massive MIMO, beamforming, and millimeter waves

Supports IoT, smart cities, autonomous vehicles, remote surgery, and AI applications

Provides speeds up to 10 Gbps with very low delay

In summary:5G evolved to deliver faster communication, support more devices, and enable future smart technologies.